On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code violation of these priciples will lead to a zero grade and is subject to horsed discipline issues

Yavuzhan Algüt

The second matrix says that if we multiply first row by -8 and add it to the second the first column entry of second row will be 0

so:

$$1.(-8) + 3\alpha = 0 \qquad \alpha = \frac{8}{3}$$

And if we multiply first row with m and add to 3rd row the first entry will be 0 so:

$$1.m + 2 = 0 \qquad m = -2$$

And we reduce the first entry of 4rd row to 0

After first elimination:

$$\begin{bmatrix} 1 & \frac{8}{3} & -y & \frac{8}{3} \\ 0 & \frac{-49}{3} & -y & \frac{-61}{3} \\ 0 & b-\frac{16}{3} & 3+2y & \frac{16}{9} \\ 0 & 34/3 & y-6 & 2-\frac{8}{3} \end{bmatrix}$$

CamScanner ile tarandı

1. devam

şimdi ilk matrixi kullanıyoruz ki 2. entry'ler 0 olsun

$\left(-\frac{49}{3}\cdot(-1)\right) + b - \frac{16}{3} = 0 \qquad b = -11$

$-\frac{49}{3}\cdot k + \frac{34}{3} = 0 \qquad k = \frac{34}{49}$

2. eliminasyon sonrası

$$\begin{bmatrix} 1 & \frac{8}{3} & -y & \frac{8}{3} \\ 0 & -\frac{49}{3} & -y & -\frac{61}{3} \\ 0 & 0 & 3+2y & \frac{16}{9} \\ 0 & 0 & y-6-yk & \frac{-2-61k}{3} \end{bmatrix}$$

3. eliminasyon 3. row'u $-1/2$ ile çarpıp 4. row'a ekliyoruz

$3+2y\cdot\left(-\frac{1}{2}\right) + y - 6 - y\cdot\frac{34}{49} = 0 \qquad y = \frac{-735}{68}$

3. eliminasyon sonucu.

$$\begin{bmatrix} 1 & \frac{8}{3} & \frac{735}{68} & \frac{8}{3} \\ 0 & -\frac{49}{3} & \frac{735}{68} & -\frac{61}{3} \\ 0 & 0 & 3+\frac{-735.2}{68} & \frac{16}{9} \\ 0 & 0 & 0 & \frac{-2-61.\frac{34}{49}}{3} \end{bmatrix}$$

upper triangular matrix

CamScanner ile tarandı

1. devam

c)

$$\begin{bmatrix} 1 & \frac{8}{3} & \frac{735}{68} & \frac{8}{3} \\ 0 & \frac{-49}{3} & \frac{735}{68} & \frac{-61}{3} \\ 0 & 0 & \frac{-1266}{68} & \frac{16}{9} \\ 0 & 0 & 0 & \frac{98-61.34}{147} \end{bmatrix} \xrightarrow[r_3\left(\frac{68}{-1266}\right)]{r_2\left(\frac{-3}{49}\right)} \begin{bmatrix} 1 & \frac{8}{3} & \frac{735}{68} & \frac{8}{3} \\ 0 & \frac{-49}{3}\cdot\frac{-3}{49} & \frac{735}{68}\cdot\frac{-3}{49} & \frac{-61}{3}\cdot\frac{-3}{49} \\ 0 & 0 & 1 & \frac{16}{9}\cdot\frac{-1266}{68} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$r_4 \cdot \frac{147}{98-61.34}$

after this we will multiply r_2 by $\frac{-3}{3}$ and add to the first row

multiply 3rd row by $\frac{735}{68} \cdot \frac{3}{49}$ and add to the r_2

CamScanner ile tarandı

2. $x = 5y$

$x - 5y = 0$ $\quad N(A) = \begin{bmatrix} 5 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$

in 3D =

$$\begin{bmatrix} 1 & -5 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$N(A)$

1 pivot 2 free variable

2D space Since Null space is 2D in 3D

if it is in 2D

$$\begin{bmatrix} 1 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$N(A)$ $\quad N(A) \begin{bmatrix} 5 \\ 1 \end{bmatrix}$

1 pivot 1 free

1D space in 2D

The other geometry can be a line or a plane inside 3D space (en az 3D space içinde olmalı yoksa 3D shape oluşturamaz),

c) $x + 4y - z = 6$ $\quad 2D \cup 2D \rightarrow 3D$

homogenious solution

$$\begin{bmatrix} -4 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$ 2D in 3D

solution matrix

$$\begin{bmatrix} 5 & 0 & -4 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

rref ↓

$$\begin{bmatrix} 1 & 0 & -4/5 & 1/5 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1/5 \end{bmatrix}$$

3 pivots means it is 3D inside 3D [number of rows]

CamScanner ile tarandı

3. a) $H = I - 2u.u^T$

$H^2 = I^2 - 2.I.2.u.u^T + 4.(u.u^T)^2$

$H^2 = I^2 - 4.u.u^T + 4.u.u^T$

$H^2 = I^2$

$H^2 = I$

$(u.u^T)^2 = u.u^T$

Since it is unit vector square of it is still same

$I^2 = I$ identity matrix

$1^2 = 1$

Unit vektör uzunluğu 1'dir

$\det(u.u^T) = 1$

b)

$\det(H) = \det(I) - 2.\det(u.u^T)$

$\det(H) = 1 - 2$

$\det(H) = -1$

it has length of $1 = |H|$

or

$H^2 = I$

$H = I$ or $H = -I$

Both is orthonormal and has length of 1

also columns are orthogonal

d) for H to be diagonalizable the eigenvector matrix determinant must be $\neq 0$ or if eigenvalues are different it is diagonalizable

CamScanner ile tarandı

c) H uzunluğu 1 dir

$$H \cdot v = u \cdot x \quad \text{eig. value}$$

uzunluğu 1

↓

unit vektör

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